

वैक्रान्त

विकृन् सति लोहानि तेन वैक्रन् क्तः स सृः

विकृन् सति रोगाणि इति वैक्रन् तः R.R.S 2/60

The one which causes Vikruti or damage to Loha or when the Vaikranta is rubbed over the surface of metals it leaves lines or scratches and can cut the thin sheets of metal hence it is called Vaikranta. Or

Vaikranta when administered in appropriate form can destroy many diseases 'Roga or Vikara' in the body and hence it is called Vaikranta.

It is also called Kuvajra or Churna Vajra as it is used as a substituent for Vajra or Diamond

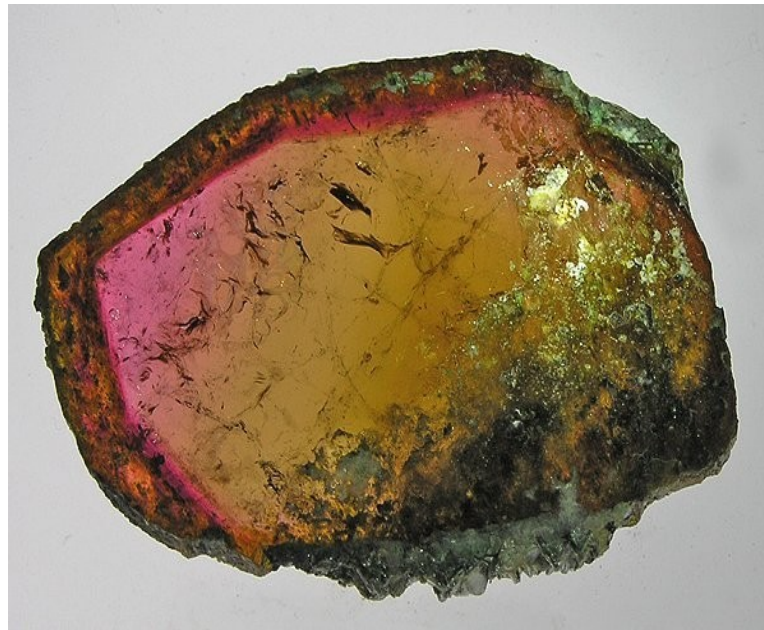
वैक्रान्तं छैव विक्रान्तं त्रैक्रान्तं तंजीर्णवज्रकम्

कुवज्रं क्षुद्रकुलिशं चूर्णवज्रञ्च तन् सप्तम् | R.T.23/155

Synonyms of Vaikranta are – Vikranta, Jeerna vajra, Kuvajra, Kshudrakulisha, Churna vajra etc



tormaline 8.webp



- **History** – 1st reference of the drug Vaikranta can be traced from Koutilyas Artha Shastra (4th cent BC) as one of the Dhatu along with other 6 metals.
- Later starting from the book Rasa Hrudaya Tantra, it has been classified under either Maharasa or Ratna varga.
- Vaikranta is the 2nd Mineral drug in Maharasa Varga. It is called Tourmaline in English and is found along with Mica.
- Tourmaline is a crystal, Borosilicate mineral compounded with elements such as Al, Fe, Mg, Na, Li and K.
- Chemically it is identified as Silicate of Mg, Mn, Fe, Ca, Na, K, Li, H, F, Al and B

- As this comes in wide variety of colors it is also classified as Semi-precious stone.
- Its important mines are situated in ‘Thoramalli’ of ShriLanka.
- Tourmaline belongs to the trigonal crystal system and occurs as long, slender to thick prismatic and columnar crystals that are usually triangular in cross section.
- It is distinguished by its 3 - sided prisms, with faces having vertical striations that produce a rounded triangular effect. (not present in other mineral)
- Tourmaline containing alkaline substances are transparent and the Iron rich Tourmalines are black to bluish black or brown in color.

- Magnesium rich varieties are brownish yellow, Lithium rich tourmalines are of different colored such as Blue, Green, Red, Yellow, Pink etc.
- Bicolored and tricolored crystals are common. Some looks like water melon, called water melon tourmaline. Some are dichroic which change color when viewed from different directions.
- **Earlier some compared Vaikranta with Fluorite or Fluorospars which is a halide mineral composed of Calcium fluoride CaF_2 .**
- **Sources** - South Africa, Canada, Peru, Mexico, Thailand, Norway, Germany, UK and in India it is found in Rajasthan and Gujarat
- **Hardness** – 4, Sp.Gr – 3.18, Refractive Index – 1.43,
- **Solubility** - Insoluble in water.

- **Mythological origin** – Vaikranta is considered as the dropped blood of Devil Mahishasura, when he was pierced by the Trishula of Lordess Durgadevi, and which was fallen in regions of Southern and northern part of Vindya hill.

Vaikranta Grahya laxana

अष्टास्रश्चाष्टफलकः षट्कोणो मसृणो गुरुः
शुद्धमिश्रितवर्णैश्च युक्तो वैक्रान्तस्तच्च स्वे | **R.R.S 2/52**

अष्टास्रमष्टफलकं सितपीतासितारुणम्
मसृणं गुरु षट्कोणं वैक्रान्तस्तच्च स्वे | **R.T.23/157**

The Vaikranta which possesses 8 edges, 8 surfaces, 6 angles, which is smooth and heavy, which is white, yellow, Black, Red or with mixed colors is considered as ideal variety.

Types of वैक्रान्त

श्वेतो रक्तश्च पीतश्च नीलः पारवतच छिः

१ मलः कृष्णश्च कर्बुरश्च अष्टधा हि सः | R.R.S 2/53

श्वेतो नीलस्य रक्तः पीतः पारवतप्रभः

ताक्ष्याभः कर्बुरः कृष्णश्चोर्णतश्चाष्टधा हि सः | R.T 23/158

Depending on the Color Vaikranta is of 8 types.

1. श्वेत वैक्रान्तं त्व(white)
2. रक्त वैक्रान्तं त्व(red)
3. पीत वैक्रान्तं त्व(yellow)
4. नीलवैक्रान्तं त्व(blue)
5. पारवतप्रभवैक्रान्तं त्व पारवतच छिः(Greyish or multicolor)
6. १ मल वैक्रान्तं त्व ताक्ष्या वैक्रान्तं त्व(Bluish or greenish)
7. कृष्ण वैक्रान्तं त्व(Black)
8. कर्बुरवैक्रान्तं त्व(blackish ash)

अशुद्ध वैक्रान्तोष

अशुद्धो वज्रवैक्रान्त तक्किलासं दाहसन् तीम्
पार्श्वपीडां तथा पान् कुकुरतस् तन्निशोधयेत् | A.P 5/168

Administration of Ashuddha Vaikranta Bhasma causes diseases such as Kilasa Kushta, Daha, Parshwapeeda and Panduroga.

वैक्रान्त त्शोधन

कुलत् क्वाथ संस्विन्नो वैक्रान्त त्तरिशुद्धे स्ति | R.R.S 2/66

- Coarsely powdered Vikranta → tied in a Pottali → Dolayantra
Swedana using Kulathakwatha → 1 yama → Vaikranta removed
washed and dried.
- Swedana with Kadalikanda Swarasa for 1 Yama.
- Nirvapa in Ashwamutra for 21 times.

वैक्रान्ताः सः सुत्रि दिनं विशुद्धाः संसृजिताः क्षारपटूनि दत्वा

अम्बुमूत्रेषु कुलत्थम् अथवा नीरेऽथवा कोद्रवपरिपक्वाः | R.R.S 2/63

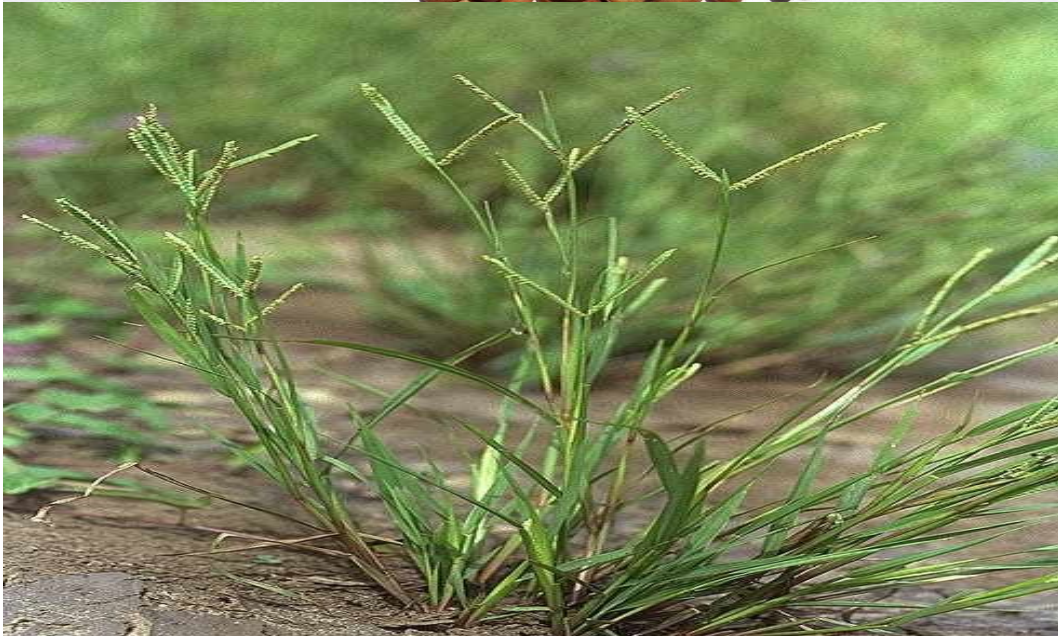
- Dolayantra Swedana of Vaikranta, using Kulattha kwatha or Nimbu swarasa or Kadalikanda swarasa or Gomutra or with Kodrava kwatha or with Kanji added with Yavakshara and Saindhava lavana.

वैक्रान्त त्सारण

1st method

समीयतेऽष्टपुटैर्गन्धमिश्रं कुद्रवसंयुतः | R.R.S 2/64

- Equal quantities of fine powders of Sh. Vaikranta and Sh. Gandhaka → taken in Khalvayantra → Nimbu swarasa → Bhavana is given → dried → Chakrikas are prepared → filled inside the Sharava Samputa → subjected to 8 Gajaputa → red colored Vaikranta bhasma



2nd Method

- Vaikranta make red hot —→ 21 times —→ powdered —→ dipped in Haya mutra (Horse urine) for Bhavana with Hayamutra —→ subjected to 8 Gajaputa.

3rd Method

- Equal quantities of Sh Vaikranta, Sh Hingula and Sh Gandhaka
Bhavana with Nimbu swarasa —→ dried —→ Chakrikas prepared
kept in Sharava Samputa —→ subjected to 8 Gajaputa.

सत घातन of वैक्रान्त

सत घातन योगेन मर्दितश्च वटीकृतः मूषास्थो

घटिका ६ सो वैक्रान्त तप्तं घृतं सृत् | A.P 5/170

Sh Vaikranta —→ triturated with Mitrapanchakas and Tankana
taken in a Musha —→ intense heat is given for 1 hour —→ white
Satva gets is collected.

Properties and Uses

वैक्रान्तः स तु त्रिदोषघ्नं षड्रसो देहदार्ढ्यकृत्
पानदूरजश्चासकासक्षयप्रमेहनुत् | A.P.5/167

- Rasa - Shadrasa
- Guna - Guru.
- Veerya - Sheeta
- Vipaka - Madhura
- Matra / Dosage – 1/24th to 1/11th Ratti (5-11mg) and according to the patient and disease condition.

Pharmacological properties

- Tridoshahara, Deha dardyakara, cures Pandu, Udara, Jwara, Swaasa, Kaasa Kshaya and Prameha rogas.

आ युःप्रदश्च बलवर्णक रोति वृष यज्ञाप्रदः सकलदोषगदापहारि
दीप्ता ग्निकृत् त्रिसमान गुणस् तस् वीक्रान्तः कः खलु वपुर्बललोहकारी
रसायनेषु सर्वेषु पूर्वगव्यः प्रतापवान्
वज्रस्थाने नियोक्तव्यो वैक्रान्तः त्सर्वदोषहाः | R.R.S 2/54,55

Vaikranta bhasma is Ayuprada, Balavarnakara, vrushya, prajnaprada, deepaka, paachaka, Dehalohakara, best Rasayana and can be administered in all diseases.

आमैक प्रयोग

1. Vaikranta Rasayana - I
Vaikranta bhasma 1 part
Svarna bhasma 1/4th part
Pippali churna 3 Parts
Vidanga churna 3 parts

- All the above ingredients mixed well with ghee and if consumed in a dose of 1 Ratti, will cure Rajayakshma, Pandu, Arsha, Shwasa, Kasa, Dushta Grahani, Urakshata, Jara etc

2. Vaikranta Rasayana - II

Parada bhasma -1 part

Neela Vaikranta Bhasma -1 part

Abhraka Satva Bhasma -1 part

All the above Bhasma mixed together and consumed in a dose of 1 Ratti with honey and ghee. It cures Chronic Grahani, Pandu, Shwasa Kasaroga etc

Important Yogas

- सूचिका भरण रस
- वसन्त कुसुमाकर रस
- क्रान्त तत्सायन
- विजयपर्पटि
- नवरत्न राजमृगाङ्कुर रस
- त्रैलोक्य यचिन्त सणि रस
- काँश्चनाभ्र रस
- It is also used for ornamental purpose, and to extract metals / elements from minerals

माक्षिक / सुवर्णमाक्षिक (COPPER PYRITE and IRON PYRITE)

Synonyms of Swarna makshika –

सुवर्णमाक्षिकं र र्णमाक्षिकं हेममाक्षिकम्
माक्षिकं माक्षिकञ्चैव ताप धातुमाक्षिकम्
माक्षिक धातुञ्च तथा तापीजञ्चापि तन् सप्तम् | R.T.21/1-2

र र्णमाक्षिक is also called सुवर्णमाक्षिक, हेममाक्षिक, माक्षिक, माक्षिक, ताप य
धातुमाक्षिक, माक्षिक धातु, तापीज etc

Synonyms of Tara makshika

रौप्यमाक्षिकमाखु स्रंतारजं तारमाक्षिकम्
विमलं श्वेतमाक्षिकं तथा रजतमाक्षिकम् | R.T.21/55

तारमाक्षिक is also called रौप्यमाक्षिक, तारज, विमल, श्वेतमाक्षिक, रजतमाक्षिक

History – The history of drug Makshika can be traced from the Samhita kaala. In Cha. Sam its therapeutic use can be seen. It was being widely used in medicinal preparations even since from Buddha period. All Rasa Granthas have dealt with this drug in detail.

Mythological origin

Makshika is considered as the blood from the foot of Lord Shri Krishna fallen on Sumeru Mountain.

It is considered as the yellowish liquid exuded from the Svarnashila mountain. As it was obtained from the banks of Tapati, it is called Tapyā or tapijā.

Nirukti

- **स्वर्णमाक्षिक** – Golden color or may be due to the presence of traces of Gold.
- **माक्षिक** – like Honey or as it melts like Madhucchista (bees wax).
- **ताप्य** – which gets dried by the Sunlight (taapa) or was collected from River bank of Tapati
- **रजतमाक्षिक** – due to the color like the Silver or due to the presence of traces of Silver in it.

माक्षिक भेद

माक्षिको द्विविधो हेममाक्षिकः समाक्षिकः

तत्राष्ट मोक्षिकः कान् कुठ लोथः र्षसन्निभः

तपतीतीरसः स्रूः पञ्चवर्णसुवर्णवत्

पाषाणबहलः प्रोक्तः स साष्ट ख्ये ल गुणात् कः | R.R.S 2/81

- According to R.R.S Makshika is of 2 types

1. Hema or Svarna makshika / Copper pyrite
2. Tara or Rajata makshika / Iron pyrite

- According to Ayurveda Prakasha 3 types

1. Hema makshika / Peeta Makshika / Copper pyrite – therapeutically more significant.
2. Tara makshika / Shweta Makshika / Iron pyrite – therapeutically less significant.
3. Kamsya makshika / Rakta Makshika / Arsenopyrite – rarely used for therapeutic purposes.

Hema makshika or Copper pyrite or Chalcopyrite (Cu FeS_2 or $\text{Cu}_2\text{S Fe}_2\text{S}_3$)

- It is obtained from Kanya kubja (Kannaouj, UP), the river bank of Tapati.
- It is widely used in Medicine.
- It belongs to Pyrite group of Minerals.
- The name pyrite in Greek, 'Puritrs' which means 'of fire' or 'in fire' .
- It usually found associated with other sulfides or oxides in quartz veins, sedimentary rock and metamorphic rock, as well as in coal beds and as a replacement mineral in fossils.
- Its color varies from golden yellow to greenish black color. Hence called '**Fools gold**'.
- Hema Makshika / Chalcopyrite : It contains 35% Copper, 30% Iron and 35% Sulphur. It has a brassy golden yellow color and a hardness of 3.5 - 4.
- It is a chief ore of Copper. Presence of Copper give it a golden color.

- Sp. Gr – 6. Luster – Metallic. Streak – Greenish black. Solubility – soluble in Nitric acid. It burns with Blue flame. It is a good conductor of Heat and electricity.
- Occurrence – Important mines are situated in US,UK, Japan, Spain Canada and Italy. In India it is available in Bihar, Rajasthan, Andra Pradesh, M.P, Sikkim, Chennai, Punjab, Darjling, Mysore etc.

Roupya / Rajata Makshika or Ironpyrite or Marcasite (Fe_2S_3 or FeS_2)

- Compared to Swarna makshika it is less used in Medicine.
- It has more impurities or sand particles and hence have less therapeutic properties.
- It is white or whitish yellow in color as it resembles Silver it is called Roupya or Rajata makshika.

- Iron pyrite is the most common of the Sulphide minerals. It is highly unstable and hence when exposed to air and water decomposes into Iron oxides, hydroxides and Sulphates.
- Hardness – 6, Sp Gr – 4.8-4.9, Fe- 46.6% and S – 53.4%

Kamsya Makshika / Arseno pyrite

- It resemble the color of Kamsya (Cu and Sn, 4:1). It is a compound of Iron pyrite with Arsenic content.

ग्राह्यलक्षण of र र्षमाक्षिक

र र्षाभं र र्षमाक्षिकं निष्ठ षो गुरुतायुतम्
का लिमां विक रेत्तत्तु करे घृष्टं न संशयः
र र्षवर्णं गुरुस्त्रिग र्षषन्नीलच छि र कुम्
क षे क नक वद् घृष्टं तद्वरं हेममाक्षिकम् | A.P 2/8

The ideal Svarnamakshika should have the following qualities.

Svarnaabha – lustre of Svarna

Nishkona – Angleless or irregular in shape

Guru – Heavy /Solid or compact

Kalima – which leaves blackish discoloration of Hands after rubbing.

Svarnavarna – Color of Svarna or Gold

Snigdha – Unctuous

Ishanneelachavi – Light bluish colored

Sphutam – Clear

Kashe Kanakavad ghrishtam – streak of the mineral also appears gold like.

ग्राह्यलक्षण रौप्यमाक्षिक

स्निग्धं क्षंकोणं गुरु च वर्तुलम् रजतोज्ज्वलम्

रौप्यं यच्चाकचिकं क्षयाज्जातं यंद्रौप्यमाक्षिकम् | R.T 21/57

- Rajata makshika which is Snigdha – Unctous, Sakona – with angles, Guru-Heavy, Vartula – Circular bodied or cubic, which possesses the shine and radiance of Silver is considered fit for therapeutic purposes.

अशुद्ध माक्षिक दोष

मन् दालत क्बलहानिमुग्रां विष्टंभितां नेत्रगदान् सकुष्ठान्
मालां विधत्तेपि च गन् धूर्वा शुद् ध्विहीनं खलु माक्षिकम् | A.P.4/11

Svarnamakshika if not properly purified and if consumed produces Mandagni, Nirbalata, Vishtambha, Netraroga, Kushta, Gandamala and Udaragata vata etc

Roupymakshika also produce the similar adverse effect like Mandagni, Kushta etc.



एरण वैललुङ्ग माक्षिद्धं शुध क्षि माक्षिकम्
सिद्धं वा कदलीकन त्रोयेन घटिका द्वयम्
तप्तं क्षिप्तं वराक्काथे शुद्धिमायाति माक्षिकम् | R.R.S 2/78

The powder of S.M is tied in a cloth like Pottali and subjected to Dolayantra swedana using either a mixture of Eranda taila and Matulunga Swarasa or in Kadalikanda swarasa for 2hrs.

- The pieces of Swarnamakshika are heated to red hot and dipped in Triphala kwatha for 7 times.
- Then the Shodhita Makshika is collected, washed, dried and stored in Airtight container.

मारण

मातुलुङ् माषुन् भ्वा यप्तिष्ठं मूषोदरे क्षिपेत्
पञ्चक्रो डपुटैर्दग् धंम्रीयते माक्षिकं खलु |R.R.S 2/79



- Sh. Makshika equal parts of Sh. Gandhaka given Bhavana → with Matulunga swarasa → chakrikas prepared and dried kept sealed in a Moosha → subjected to 5 Varaha Puta Raktotpala (red lotus) colored Swarna makshika bhasma is obtained.
- S.M Bhasma can also be prepared by triturating with Nimbu swarasa and subjecting to 10 Varaha Puta.
- Rajata Makshika Bhasma can also be prepared with the same methodologies.

Properties of सुवर्णमाक्षिक भस्म

सुवर्णमाक्षिकं वृष्यं यंधुरं तु रसायनम्
तिक्तं सुवर्णमाक्षिकं चक्षुष्यं यत्रिदोषघ्नं परं मतम् ।
क्षयमर्शांसि मेहानि विविधा बस्तिवेदनाः
पाण्डुपित्तकुष्ठानि विषदोषादिकान् हरेत् ।
जीर्णजं वमपित्तं सामन् बालमरोचकम्
अनिद्रां नाशयत् सुयोगवाहि परं मतम् ॥R.T.21/26-28

Rasa – Madhura, Tikta rasa

Guna – Yogavahi

Dose : ½ to 2 Ratti,

- Anupaana : Triphala churna, Trikatu churna, Vidanga churna, Ghruta and Madhu
- Chemical composition – Copper, Iron, Sulfur, Copper Iron sulfide, Silica, Zinc, Manganese, Lead etc.
- Apathya – Kakamachi, Kapota mamsa, Kulatha.
- Aushadha sevana kaala - 2 times a day.
- **Pharmacological properties** – Properly prepared Swarnamakshika bhasma is Vrishya, Rasayana, Svarya, Chakshushya, tridoshahara, Kshaya, Arsha, meha, Vastivedana, Pandu, swayathu, Kushta, Vishadosha, Jeernajwara, Apasmara, Agnimandya, Arochaka, and Anidra

अशुद्धं माक्षिकं दोष

अशुद्धं माक्षिकं कुर्यादान् क्षुण्णं क्षयं क्रिमीन् । रसराजसुन् द

Consumption of impure or improperly prepared Makshika bhasma causes Andhatva, Kushta, Khsayaroga and Krimi roga.

Antidote : For the consumption of Apakwa Swarna Makshika Bhasma one should consume Kulattha kwatha or Dadima twak kwatha.

Avasthanusara uses –

- Rajayakshma – SM Bhasma + Sh Shilajatu + Abhraka churna + Vidanga churna + Madhu
- Muthra krucchra – SM Bhasma + Yavakshaara

- Shotha – SM Bhasma + Puarnava+Shunti+ Guduchi+ shunti + Darvi kwatha .
- Sheeta jwara – S.M Bhasma + Atasi beeja + Karanja beeja
- Pittaja unmada and nidra nasha – Chandra puti Pravala Bhasma and Dugdha

Pharmacological actions –

- Hepatoprotective – improves liver functions
- Antioxidants – scavenging effect, delays aging and cellular damage.
- Anti inflammatory – hence useful in gastritis and arthritis.
- Antimicrobial - strengthens the immune system
- Antidiabetic – improve insulin sensitivity

- Anti arthritic – by reducing inflammation
- Nervine tonic – strengthens the nervous system
- Cardioprotective – regulates BP and heart condition
- Rejuvenate – enhances vitality and promote longevity.

Important योग s

1. चन् ब्रभा वटी
2. रत्नगर्भ पोदुली
3. ग्रहणीक पाट रस
4. जयमङ्गल रस
5. त्रैलोक्य चिन्ता सानि रस
6. पञ्चामृत पर्पटी

सट षातन

त्रिंशांश नागसंयुक्तं क्षारैरम् श्लै मर्दितम् ।

६ सांप्रकटमूषायां सट वंश्वति माक्षिकम् । R.R.S 2/81

To the 1/30th part of melted Naga powders, 1part of Sh Makshika is added little by little till Naga becomes powder. To this 3 Kshara churna are added and triturated with Amla dravya Swarasa, paste is prepared, kept in Moosha and blow in Satvapatana Koshti. In this way Makshika satva is extracted.

The Satva thus obtained is melted and poured into Nirgundi swarasa for 7 times to eradicate the Naga which is present with the Makshika satva.

पादांश टङ्कोपेतं मर्दितं हेममाक्षिकम्

६ सामूषागतं सत वंशत श्चा सुनिश्चितम् | R.T.21/46

1 part of Sh Svarnamakshika is triturated with 1/4th part of Sh Tankana, the mixture is kept in Moosha heated intensely till the copper like Makshika satva is extracted.

माक्षिकसत क्लक्षण

गुञ्जाबीजसमच च्छाद्रुतद्रावं च शीतलम्

ताप सत वविशुद्धं तद्देहलोहकरं परम् | R.T.S 2/84

- The Makshika satva looks red like Gunjabeeja, melts quickly, feel cold to touch and is capable of making the body strong like a metal.

5. Research updates –

1. Analytical study of raw swarnamakshika and its Bhasma through TEM and EDAX by Sudhaldev Mahapatra - revealed Iron, Copper and Sulfur in major quantity and K, Mg, Al, Si in trace level. TEM shown significant reduction in grain size from 5-10microns in Raw SM to 50-200 nm in SM Bhasma.
2. A review on therapeutic application of Swarnamakshika Bhasma by dr Shruti Nambiar, concluded that the SM Bhasma having nano particle and inherent properties can be prescribed for many conditions with appropriate Dosage and Anupana.